Chapter 7: Droplet-Borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

1. **Describe the causative agents and transmission routes of major droplet-spread diseases.**
2. **Recognize the clinical features and complications of each disease.**
3. **Understand the epidemiological patterns and risk groups.**
4. **Apply prevention and control strategies, including vaccination and outbreak response.**
5. **Interpret surveillance data and contribute to public health interventions.**

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1. Introduction

Droplet-borne diseases are transmitted through respiratory droplets expelled during coughing, sneezing, talking, or certain medical procedures. These droplets typically travel short distances (<1 meter) and deposit on mucous membranes of nearby individuals. Such diseases are highly contagious and pose significant public health challenges, especially in crowded or healthcare settings.

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2. Overview of Transmission

Feature	Droplet Transmission
Particle size	>5 microns
Distance of spread	Usually <1 meter
Mode of entry	Mucous membranes (eyes, nose, mouth)
Environmental survival	Limited
Prevention	Masks, distancing, vaccination, hygiene

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3. Key Droplet-Borne Diseases

3.1 Influenza

Causative Agent:
Influenza virus (Types A, B, C); Type A causes pandemics.

Transmission:
Respiratory droplets, contaminated surfaces, close contact.

Incubation Period:
1–4 days

Clinical Features:

- **Sudden onset of fever, chills, headache**
- **Myalgia, sore throat, dry cough**
- **Fatigue, sometimes gastrointestinal symptoms in children**

Complications:

- **Pneumonia (viral or secondary bacterial)**
- **Exacerbation of chronic diseases**
- **Death in high-risk groups**

Prevention:

- **Annual influenza vaccination**
- **Respiratory hygiene and handwashing**
- **Antiviral prophylaxis in outbreaks**

Epidemiological Notes:

- **Seasonal peaks in winter**
- **High-risk groups: elderly, pregnant women, healthcare workers**
- **Surveillance through sentinel sites and virological testing**

3.2 Measles

Causative Agent:
Measles virus (Paramyxovirus family)

Transmission:
Highly contagious via respiratory droplets and airborne spread

Incubation Period:
7–14 days

Clinical Features:

- **High fever, cough, coryza, conjunctivitis**
- **Koplik spots (white lesions on buccal mucosa)**
- **Maculopapular rash starting on face and spreading downward**

Complications:

- **Otitis media, pneumonia, encephalitis**
- **Subacute sclerosing panencephalitis (rare, late complication)**

Prevention:

- **Measles vaccine (part of MMR)**
- **Isolation of cases**
- **Mass immunization campaigns during outbreaks**

Epidemiological Notes:

- **Endemic in areas with low vaccine coverage**
- **Outbreaks common in refugee camps, schools**
- **WHO targets measles elimination through ≥95% vaccine coverage**

3.3 Mumps

Causative Agent:

Mumps virus (Paramyxovirus family)

Transmission:

Respiratory droplets, direct contact with saliva

Incubation Period:

16–18 days

Clinical Features:

- **Fever, headache, malaise**
- **Painful swelling of parotid glands (unilateral or bilateral)**
- **Difficulty chewing or swallowing**

Complications:

- **Orchitis (in post-pubertal males), oophoritis**
- **Meningitis, pancreatitis, deafness**

Prevention:

- **Mumps vaccine (part of MMR)**
- **Isolation of cases for 5 days after onset of parotitis**

Epidemiological Notes:

- **Occurs in unvaccinated populations**
- **Outbreaks in universities, military camps**
- **Vaccine coverage reduces incidence significantly**

3.4 Rubella

Causative Agent:

Rubella virus (Togavirus family)

Transmission:

Respiratory droplets; vertical transmission during pregnancy

Incubation Period:

14–21 days

Clinical Features:

- **Mild fever, lymphadenopathy (especially postauricular)**
- **Fine pink rash starting on face and spreading**
- **Often asymptomatic in children**

Complications:

- **Congenital Rubella Syndrome (CRS): cataracts, deafness, heart defects**
- **Arthralgia in adults**

Prevention:

- **Rubella vaccine (part of MMR)**
- **Vaccination of women of childbearing age**
- **Surveillance of CRS cases**

Epidemiological Notes:

- **CRS is preventable through immunization**
- **Rubella elimination is a global goal**
- **Serological screening recommended before pregnancy**

3.5 Whooping Cough (Pertussis)

Causative Agent:

Bordetella pertussis **(Gram-negative coccobacillus)**

Transmission:

Respiratory droplets; highly contagious

Incubation Period:

7–10 days

Clinical Features:

- **Catarrhal phase: mild cough, rhinorrhea**
- **Paroxysmal phase: severe coughing fits with inspiratory “whoop”**
- **Post-tussive vomiting, cyanosis in infants**

Complications:

- **Pneumonia, seizures, encephalopathy**
- **Death in infants <6 months**

Prevention:

- **DTaP vaccine in childhood; Tdap booster in adolescence/adulthood**
- **Cocooning strategy for newborns**
- **Antibiotic prophylaxis for close contacts**

Epidemiological Notes:

- **Resurgence due to waning immunity**
- **Surveillance essential for early detection**
- **Vaccine coverage critical for infant protection**

3.6 Meningococcal Meningitis

Causative Agent:

***Neisseria meningitidis* (serogroups A, B, C, W, Y)**

Transmission:

Respiratory droplets; close contact in crowded settings

Incubation Period:

2–10 days

Clinical Features:

- **Sudden fever, headache, neck stiffness**
- **Photophobia, altered consciousness**
- **Petechial or purpuric rash (invasive disease)**

Complications:

- **Septicemia, shock, death**
- **Neurological sequelae: hearing loss, cognitive impairment**

Prevention:

- **Meningococcal vaccines (conjugate and polysaccharide)**
- **Chemoprophylaxis for close contacts (e.g., rifampicin)**
- **Isolation of cases and outbreak control**

Epidemiological Notes:

- **Epidemics in “meningitis belt” of sub-Saharan Africa**

- **Surveillance and mass vaccination campaigns essential**
- **School and dormitory outbreaks require rapid response**

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4. Summary Table

Disease	Agent	Incubation	Key Symptoms	Prevention
Influenza	Influenza virus	1–4 days	Fever, cough, myalgia	Annual vaccine
Measles	Measles virus	7–14 days	Fever, rash, Koplik spots	MMR vaccine
Mumps	Mumps virus	16–18 days	Parotid swelling, fever	MMR vaccine
Rubella	Rubella virus	14–21 days	Rash, lymphadenopathy	MMR vaccine
Pertussis	*B. pertussis*	7–10 days	Whooping cough, vomiting	DTaP/Tdap vaccine
Meningitis	*N. meningitidis*	2–10 days	Fever, neck stiffness, rash	Meningococcal vaccine

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1. **Measles can be transmitted through airborne particles as well as respiratory droplets.**
   True
2. **Rubella infection during pregnancy poses no risk to the fetus.**

False

3. **Mumps primarily affects the liver and causes jaundice.**

False

4. **Annual vaccination is recommended to prevent seasonal influenza.**
   True
5. **Pertussis is most dangerous in infants under six months of age.**
   True